



9042 - For the Love of Exoplanets: Characterization of the Internal Structure of Hot-Jupiters

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Although we have learned much about the atmospheres of hot-Jupiters over the past two decades, we are still largely ignorant of the structure and composition of their interiors. This is because these highly-irradiated worlds experience radius inflation, and consequently their position on a mass-radius diagram is not purely a function of their structure. If we could overcome this challenge, however, constraints on their core mass fractions would be extremely valuable and might inform an unresolved tension. Although planetary formation theories suggest gas giants should contain cores of about 15 Earth masses, recent work suggests that less-irradiated giant planets may possess cores of up to 100 Earth masses. New measurements of a poorly-sampled sub-population, hot-Jupiters, could help determine whether heavy cores are in fact common beyond the solar system. We propose produce six of these measurements through a re-analysis of archival JWST transit data with a focus on fitting their deformations from spherical symmetry.

OBSERVING DESCRIPTION